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OCR GCSE Geography B (J384) — Paper 1: Our Natural World · Topic 2: Changing Climate

1.3 Changing Climate

1. Defining Climate Change

Weather — the short-term atmospheric conditions at a specific time and place (temperature, rainfall, wind speed, cloud cover). Can change hour by hour.

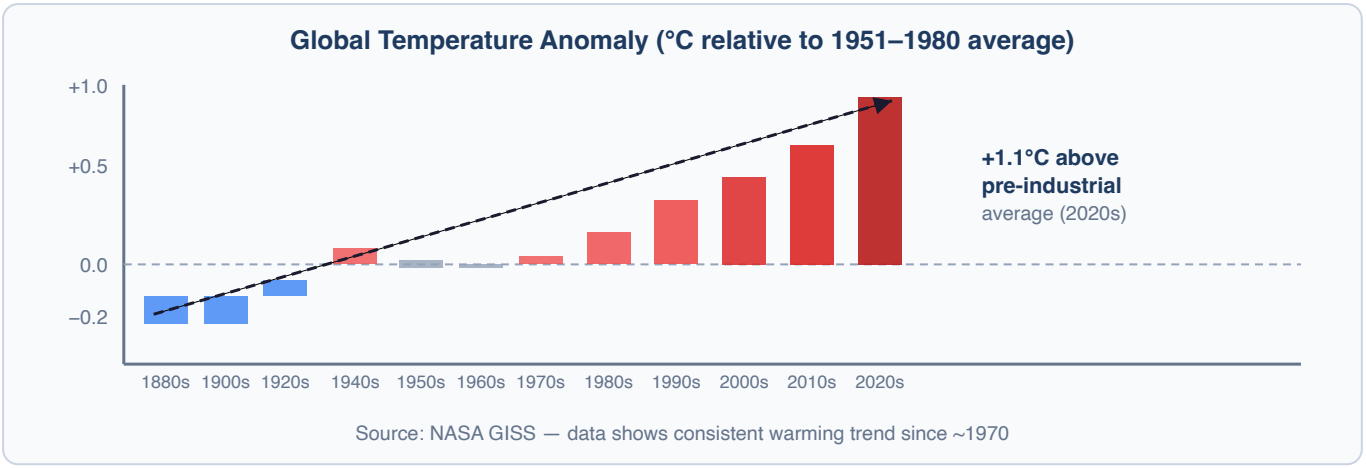
Climate — the long-term average pattern of weather conditions for a region, calculated over at least 30 years. Changes slowly over decades or millennia.

Climate Change — a long-term shift in global average temperatures and precipitation patterns, beyond natural variability. The current concern is the *unprecedented rate of warming since industrialisation (~1750)* driven largely by human activities.

The Intergovernmental Panel on Climate Change (IPCC) — the UN scientific body that reviews global climate research — concluded in its 2021 report (AR6): *"It is unequivocal that human influence has warmed the atmosphere, ocean and land."*

2. Evidence of Climate Change

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The Quaternary period — the most recent geological period, covering roughly the last **2.6 million years** to the present day. During the Quaternary, the global climate has swung between cold **glacial periods** (ice ages, when ice sheets covered much of Britain) and warmer **interglacial periods** such as the one we are in now, which began about 11,700 years ago. Evidence of these past changes tells us how the climate behaves naturally — and why the recent rate of warming is so unusual.

Key Evidence

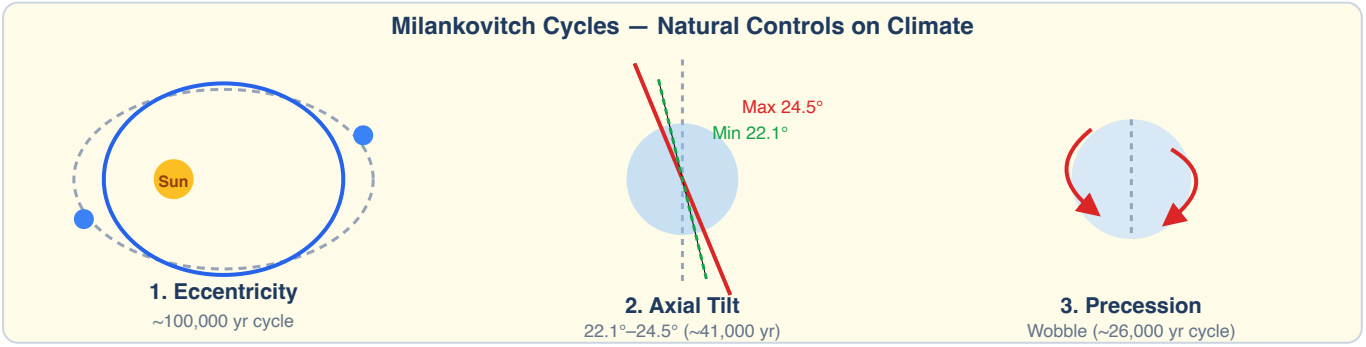
Evidence Type	Detail	What it shows
Instrumental temperature records	NASA/NOAA data: global average +1.1°C above pre-industrial (2011–2020 average)	Consistent warming trend; the ten hottest years on record were all 2015–2024 (2024 the hottest)
Ice cores (EPICA Dome C and Vostok, Antarctica)	Air bubbles trap ancient atmosphere; 800,000 years of CO_2 and temperature data	Strong correlation between CO_2 and temperature; current CO_2 levels (~422 ppm, 2024) are highest in 800,000 years
Dendrochronology	Tree ring width and density reveal past climate — wider rings = warmer, wetter years	Medieval Warm Period (~900–1300), Little Ice Age (~1300–1850) visible in record

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Glacier retreat	Rhône glacier (Switzerland): retreated well over 1km since the mid-1800s. Arctic sea ice extent declining 13% per decade since satellite records began (1979)	Widespread global warming; positive feedback as less ice = less albedo
Sea level rise	+20cm since 1900; averaging ~3.4mm/year since 1993 and now over 4mm/year (satellite altimetry)	Thermal expansion of warming oceans + meltwater from glaciers and ice sheets
Changing seasons	Cherry blossom in Japan now peaks ~10 days earlier than in 1950; UK spring arriving earlier	Plants and animals responding to warmer temperatures
Species range shifts	Many UK butterflies (e.g. comma, orange-tip) have spread far northwards since the 1970s; tropical fish appearing in North Sea	Species tracking warming climates northward/to higher altitudes

Exam Tip: When asked for evidence of climate change, use specific data (figures, dates, measurements) — not vague statements like "glaciers are melting." Examiners award marks for named examples with statistics. E.g. "Arctic sea ice extent has declined by 13% per decade since 1979 according to NASA satellite data."

3. Natural Causes of Climate Change



Milankovitch Cycles

Cyclical changes in Earth's orbit and orientation relative to the Sun drive natural climate cycles over tens of thousands of years:

- **Eccentricity** (~100,000 years) — shape of Earth's orbit changes from nearly circular to more elliptical. When more elliptical, seasonal variation in solar energy received is greater.
- **Axial tilt** (~41,000 years) — angle of Earth's axis varies from 22.1° to 24.5°. Greater tilt = more extreme seasons (hotter summers, colder winters). Current tilt is 23.5° and slowly decreasing.
- **Precession** (~26,000 years) — Earth's axis wobbles like a spinning top. Changes which hemisphere experiences summer closest to the Sun, altering which seasons are more intense.

Together, these cycles explain the glacial and interglacial periods (ice ages) of the past 2.6 million years. However, they **cannot explain the rapid warming since 1750** — Milankovitch cycles operate over far longer timescales and current orbital forcing actually points toward slight cooling over the next few thousand years.

Solar Variation

The Sun's energy output is not constant. It varies by approximately **0.1% over an 11-year sunspot cycle**. Periods of high sunspot activity correspond to slightly higher solar output. Scientists have identified longer periods of very low solar activity — the **Maunder Minimum (1645–1715)** — which coincides with the coldest phase of the Little Ice Age. However, solar output has been *flat or slightly declining* since the 1980s, while global temperatures have continued to rise sharply. This rules out solar variation as a primary driver of current warming. The IPCC (2021) estimates that natural factors (solar and volcanic) changed global temperature by only about −0.1 to +0.1°C since 1850–1900, whereas greenhouse gases caused roughly 1.5°C of warming, partly offset by cooling from aerosol pollution.

Volcanic Eruptions

Large explosive eruptions inject sulphur dioxide (SO_2) into the stratosphere. This forms **sulphate aerosols** — tiny droplets that reflect incoming solar radiation back to space, causing short-term surface cooling. Key examples:

- **Mt Pinatubo, Philippines (1991)** — second-largest eruption of the 20th century; injected 20 million tonnes of SO_2 into the stratosphere. Global average temperature fell ~0.5°C for 1–2 years before aerosols settled.
- **Laki, Iceland (1783)** — fissure eruption releasing enormous SO_2 clouds; caused harvest failures across Europe and may have contributed to famine in Egypt by disrupting the Nile flood.

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- **Tambora, Indonesia (1815)** — the largest eruption in recorded history; caused 1816 to be known as the "Year Without a Summer" across the Northern Hemisphere; the eruption and its aftermath killed an estimated 71,000+ people in Indonesia, and 1816 crop failures caused famine in Europe and elsewhere.

Volcanic eruptions produce *cooling*, not warming, and effects are temporary (1–3 years). They cannot account for the sustained warming trend observed since 1750. Some volcanoes also emit CO_2, but human emissions are now estimated to be **40–100 times greater** than all volcanic CO_2 output combined.

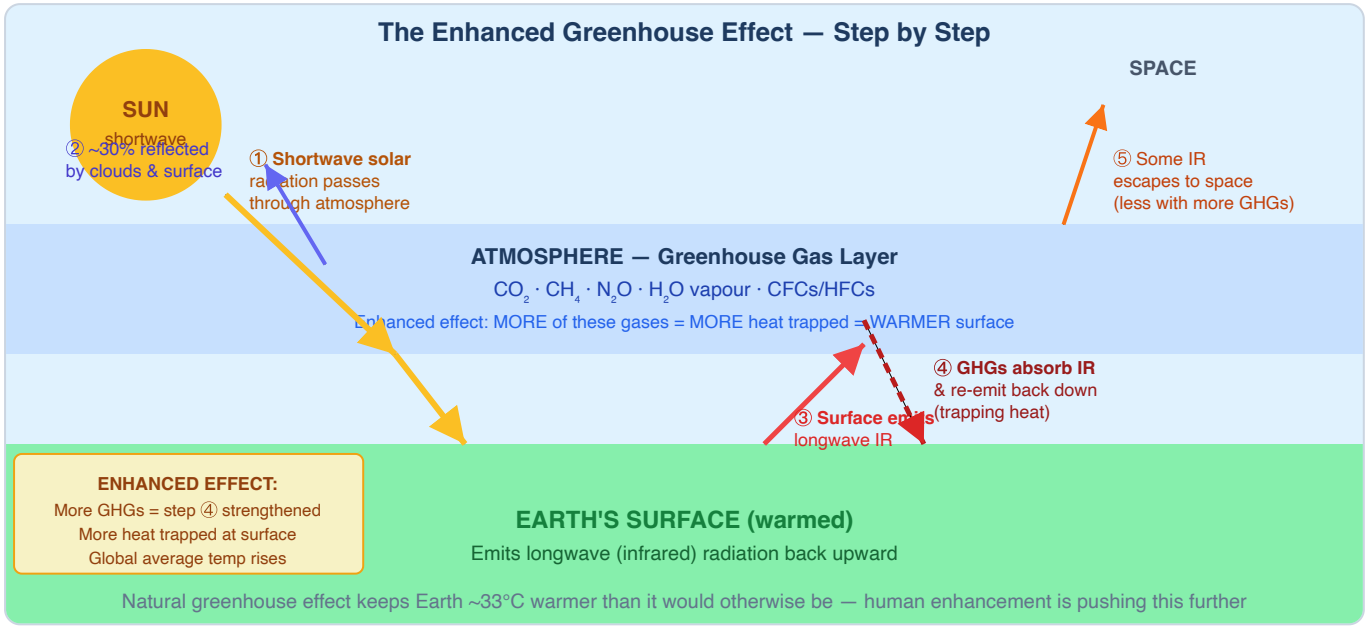
Natural vs. Human Causes — Comparison

Factor	Timescale of effect	Direction of effect	Can explain post-1750 warming?
Milankovitch cycles	Tens of thousands of years	Gradual warming/cooling	No — too slow; current orbit points to mild cooling
Solar variation	11-year cycles / centuries	Minor warming/cooling	No — solar output flat/declining since 1980s
Volcanic eruptions	1–3 years	Short-term cooling	No — cause cooling, not warming; too brief
Human GHG emissions	Decades–centuries	Warming	Yes — correlates precisely with industrialisation and CO_2 rise

Exam Tip: A common exam question asks you to distinguish between natural and human causes of climate change, then explain why scientists believe human causes are dominant. Key arguments: (1) the rate of warming is far faster than any natural cycle — over 1°C in about 150 years vs. roughly 4–7°C over about 10,000 years at the end of the last ice age; (2) ice core data shows CO_2 levels are unprecedented in 800,000 years; (3) solar output has not increased but temperature has; (4) isotopic "fingerprinting" of CO_2 in the atmosphere confirms it comes from fossil fuels (different carbon isotope ratio to volcanic CO_2).

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4. Human Causes — The Enhanced Greenhouse Effect



Natural Greenhouse Effect — essential for life on Earth. Without it, average global temperature would be −18°C (instead of current +15°C). GHGs (CO_2, water vapour, methane) in the atmosphere trap some of the longwave infrared radiation emitted by Earth's surface after it absorbs incoming solar energy.

Enhanced Greenhouse Effect — human activities increasing the concentration of GHGs in the atmosphere. More GHG molecules means more infrared radiation is absorbed and re-emitted back toward Earth's surface before it can escape to space, causing global warming. The mechanism: GHG molecules vibrate when they absorb infrared energy, then release that energy in all directions — including back downward.

Key Greenhouse Gases and Their Sources

GHG	Main human sources	Global Warming Potential (GWP)	% of human GHG emissions
Carbon dioxide (CO_2)	Burning fossil fuels (coal, oil, gas); deforestation (less absorption, carbon released)	1 (reference gas)	~76%

Methane (CH_4)	Livestock (enteric fermentation — livestock cause ~14.5% of all human GHG emissions, FAO); rice paddies; landfill; gas leaks; permafrost thaw	80x more potent than CO_2 over 20 years	~16%
Nitrous oxide (N_2O)	Agricultural fertilisers (nitrogen-based); livestock waste; combustion	265x CO_2 over 100 years	~6%
CFCs & HFCs	Refrigerants, aerosols (CFCs phased out by the Montreal Protocol; HFCs being phased down under its 2016 Kigali Amendment); industrial gases	Thousands × CO_2	~2%

Current CO_2 levels: ~422 ppm (parts per million) in 2024 — the highest in at least 800,000 years (EPICA Dome C ice core data). Pre-industrial level was ~280 ppm. Rate of increase is ~2.5 ppm per year — faster than any natural rate seen in the ice core record.

Role of Deforestation

Forests act as **carbon sinks** — absorbing CO_2 from the atmosphere through photosynthesis. Deforestation contributes ~10% of global GHG emissions in two ways:

1. Burning or decomposing trees releases stored carbon as CO_2.
2. Fewer trees means less CO_2 is absorbed — reducing the natural sink capacity.

Amazon deforestation is a major concern — roughly 17–20% of the Amazon has been cleared since 1970. Some scientists warn it may reach a "tipping point" where it switches from a carbon sink to a carbon source. Parts of the south-eastern Amazon already emit more CO_2 than they absorb.

5. Impacts of Climate Change

The impacts of climate change are wide-ranging, affecting economies, societies, and natural environments. They are experienced unequally — Low Income Countries (LIDCs), small island states, and communities in the Global South face the greatest risks despite contributing the least to emissions.

Economic Impacts

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- **Agriculture losses** — changing rainfall patterns, more frequent droughts and floods, and extreme heat reduce crop yields. Sub-Saharan Africa crop yields predicted to fall 8–22% by 2050 (IPCC). Coffee production in Ethiopia — one of the world's leading exporters — faces a 40–59% reduction in suitable growing area by 2100.
- **Extreme weather damage costs** — insurance industry losses from extreme weather have grown dramatically. Munich Re estimated around $320 billion in economic losses from natural disasters in 2024, with extreme weather events accounting for the majority. The Caribbean economy loses an estimated 1–2% of GDP annually to hurricane damage — a figure set to rise.
- **Infrastructure damage** — flooding, storm surges, and permafrost thaw damage roads, buildings, and pipelines. In Alaska, permafrost thaw is causing roads and buildings to subside; estimated repair costs run to billions of dollars. UK flooding in winter 2019–2020 caused hundreds of millions of pounds of damage.
- **Tourism losses** — the Great Barrier Reef contributes AUD$6.4 billion to the Australian economy annually; coral bleaching threatens this. Alpine ski resorts losing viability as snowfall decreases — many lower-altitude Swiss resorts could become unviable by 2100 under high-emission scenarios.
- **Fisheries decline** — ocean warming and acidification disrupt fish stocks. The North Sea cod fishery has declined by 80% since 1970, partly due to warming waters reducing food availability for cod larvae.
- **Energy costs** — increased cooling demand (air conditioning) and reduced efficiency of fossil fuel power plants in extreme heat raise energy costs. US electricity demand for cooling is projected to rise significantly by 2050.

Social Impacts

- **Food insecurity** — changing monsoons threaten rice and wheat harvests for 2 billion people in South Asia. Some estimates suggest climate change could push up to 183 million additional people into hunger by 2050.
- **Water stress** — glacial retreat in the Himalayas threatens freshwater supply for over 1.9 billion people who depend on rivers fed by glacial meltwater (Ganges, Indus, Yangtze). Short-term increase in meltwater is followed by long-term reduction as glaciers disappear.
- **Health impacts** — malaria mosquitoes (Anopheles) moving to higher altitudes in Africa and Asia as temperatures rise; now found in Ethiopian Highlands where the population has no immunity. Heatwave mortality: European heatwave (2003) killed ~70,000 people; UK heatwave (July 2022,

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40.3°C — the first time 40°C was recorded) caused an estimated 3,000 excess deaths. Flood-driven cholera outbreaks are increasing (e.g. Cyclone Idai, Mozambique, 2019).

- **Climate refugees and displacement** — World Bank estimates 216 million internal climate migrants by 2050. Bangladesh: 17 million people could be displaced by a 1m sea level rise. Kiribati and Tuvalu face complete inundation. Syria's 2006–2010 drought (linked to climate change) contributed to internal displacement that preceded the Syrian conflict — an example of climate change acting as a "threat multiplier."
- **Mental health** — "eco-anxiety," trauma after extreme weather events, and loss of cultural identity for communities displaced from ancestral lands are growing concerns, particularly among young people.

Environmental Impacts

- **Arctic sea ice loss** — Arctic sea ice extent has declined by 13% per decade since 1979. The Arctic is warming 4x faster than the global average due to the albedo feedback loop (see Section 4 diagram).
- **Ocean acidification** — oceans absorb ~26% of human CO_2 emissions; CO_2 dissolves to form carbonic acid. Ocean pH has fallen from 8.2 to 8.1 since pre-industrial times — a 30% increase in acidity. This weakens the shells of molluscs and corals. Great Barrier Reef has lost 50% of its coral cover since 1995; 2016 and 2017 bleaching events affected 91% of reefs surveyed.
- **Permafrost thaw** — Arctic permafrost stores twice as much carbon as is currently in the atmosphere. Thawing releases methane (a powerful GHG), creating a dangerous positive feedback loop. Siberia's permafrost is thawing at an accelerating rate; sinkholes called "thermokarst" are appearing across Siberia.
- **Ecosystem disruption and biodiversity loss** — IPCC (2022): at 3°C of warming, roughly 12% (possibly up to ~29%) of land species face a very high risk of extinction. Polar bears losing hunting habitat as sea ice retreats earlier each spring; body condition scores declining (thinner bears, fewer cubs). Coral triangle (SE Asia) supports 75% of world's coral species and 120 million people — severely threatened. Phenological mismatch — flowers blooming and insects emerging out of sync disrupts pollination.
- **Sea level rise** — thermal expansion (40%) + land ice melt (60%). IPCC projects 0.28–1.01m of sea level rise by 2100 depending on emissions. Beyond 2100, even if warming stabilises, seas will continue to rise for centuries. Greenland alone holds enough ice to raise global sea levels by 7m if completely melted.

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Exam Tip: Climate change impacts are *geographically unequal* — LIDCs are most vulnerable despite contributing least to the problem. This "climate justice" argument is key for evaluation questions. AC carbon emissions cause disproportionate harm to LIDC populations who lack resources to adapt. Always link economic, social, and environmental impacts together — they are interconnected (e.g. coral reef loss = environmental impact + economic impact on tourism and fisheries + social impact on food security for coastal communities).

6. Case Studies

Case Study: Arctic Sea Ice Loss — Impacts on Wildlife and Indigenous Communities

Location & Scale: The Arctic Ocean and surrounding tundra, including northern Canada, Alaska, Russia, Greenland, and Scandinavia. The Arctic is warming at 3–4 times the global average rate, making it the fastest-warming region on Earth.

Scale of Ice Loss:

- Arctic sea ice extent has declined by approximately **13% per decade** since satellite records began in 1979 (NASA).
- Summer sea ice volume has declined by over **75%** since the 1980s.
- September 2012 saw the lowest Arctic sea ice extent ever recorded — 3.41 million km² (compared to ~7 million km² in the 1980s).
- Scientists predict the Arctic Ocean could be essentially **ice-free in summer by the 2050s** under current emission trends.

Impacts on Wildlife:

- **Polar bears (Ursus maritimus)** — depend on sea ice as a hunting platform to catch ringed seals. With ice forming later in autumn and breaking up earlier in spring, polar bears have a shorter hunting season. Western Hudson Bay polar bear population has **roughly halved since the 1980s**. Bears are coming ashore thinner and in poorer body condition; cub survival rates are falling. The IUCN lists polar bears as Vulnerable; some models predict a 30% population decline by 2050.
- **Walruses** — use sea ice as a resting platform between foraging dives. Without ice, tens of thousands are forced ashore at once (a "haul-out"), causing deadly stampedes. In 2014, about

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35,000 walruses hauled out near Point Lay, Alaska; such crowding can cause stampedes that kill calves.

- **Narwhals, beluga whales, and ringed seals** — habitat and food sources disrupted. Killer whales are moving into newly ice-free Arctic waters, increasing predation on narwhals that were previously protected by sea ice.
- **Arctic cod** — a key food source in the Arctic food web; its habitat is under pressure from warming and acidifying waters. Decline cascades through the food chain to seals, bears, and seabirds.

Impacts on Indigenous Communities:

- **Inuit peoples** (Canada, Greenland, Alaska) have lived in the Arctic for thousands of years, developing deep cultural and practical knowledge of sea ice. An estimated **180,000 Inuit** live in the Arctic.
- **Traditional hunting** — Inuit hunt ringed seals, polar bears, walruses, and narwhals from sea ice. Unpredictable ice conditions make hunting more dangerous; hunters are falling through thinner ice more frequently, and search-and-rescue call-outs have increased in some communities as ice that was reliable for generations becomes unpredictable.
- **Food security** — country food (traditional food obtained through hunting, fishing, foraging) provides a large share of the diet in many remote Inuit communities and is central to cultural identity. Declining wildlife availability is driving food insecurity; studies show increasing rates of food insecurity in Inuit communities.
- **Permafrost thaw** — Inuit communities built on permafrost are seeing homes, roads, and water infrastructure sink and crack. Shishmaref, Alaska (600 Inuit residents) is losing its coastline rapidly to erosion; residents voted to relocate in 2016, at an estimated cost of around $180 million.
- **Cultural and spiritual loss** — Inuit language has over 50 words for sea ice; entire vocabulary and knowledge systems tied to ice conditions are becoming obsolete. Elders describe a world that has become unrecognisable. The Inuit Circumpolar Council has submitted a petition (2005) to the Inter-American Commission on Human Rights, arguing that climate change violates their fundamental rights.
- **Positive economic impacts (contested)** — the opening of the Northwest Passage as a shipping route and access to Arctic resources brings economic opportunity but also threatens the region with pollution and industrial development, compounding environmental pressures.

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Conclusion: Arctic sea ice loss is a powerful example of how climate change disproportionately affects communities — particularly indigenous peoples — who have contributed least to GHG emissions. It also illustrates the concept of positive feedback loops: less ice → lower albedo → more absorption → more warming → less ice.

Case Study: Maldives and Pacific Islands — Sea Level Rise Threat

Location: The Maldives, an island nation in the Indian Ocean comprising 1,200 islands and 26 atolls; total population ~540,000. Also relevant: Kiribati, Tuvalu, Marshall Islands, and Funafuti (Tuvalu's capital, the most densely populated area anywhere in the Pacific).

The Physical Threat:

- The Maldives has an average elevation of just **1.5 metres above sea level** — 80% of land is less than 1 metre above sea level. It is one of the lowest-lying nations on Earth.
- Sea level around the Maldives has risen by roughly **3–4mm per year** since the 1990s, and the rate is accelerating.
- IPCC projections: under a medium-emissions scenario, sea level could rise 0.4–0.6m by 2100 — enough to render most of the Maldives uninhabitable through storm surge, saltwater intrusion of freshwater, and coastal erosion even without permanent inundation.
- High-tide and storm-surge flooding events are becoming more frequent on some atolls.
- Tuvalu (area: 26 km²; 11,000 people) — storm surge events now regularly flood homes and contaminate the freshwater lens (thin layer of fresh groundwater). The country's highest natural point is 4.6m. In 2021, Tuvalu's foreign minister delivered a speech to COP26 standing knee-deep in seawater to illustrate the threat.

Social and Economic Impacts:

- **Tourism industry under threat** — tourism accounts for 28% of Maldives' GDP. Coral reef bleaching (driven by ocean warming and acidification) is destroying the marine ecosystem that attracts visitors. Nearly all coral in the Maldives bleached in 1998 and again in 2016.
- **Freshwater crisis** — saltwater intrusion into groundwater supplies contaminates the freshwater lens on which atolls depend. The Maldives now relies heavily on desalination — expensive and energy-intensive.

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- **Forced relocation planning** — in 2008 the Maldives' president proposed a fund to buy land abroad (e.g. in Sri Lanka, India or Australia) as a potential homeland of last resort. The government has announced plans to raise artificial islands (like Hulhumalé) and reclaim land to create higher ground.
- **Kiribati** has purchased 6,000 acres of land in Fiji (the "Kiribati Purchase") which has been discussed as a possible relocation site for its ~120,000 citizens, although it is currently used mainly for food security.
- **Climate justice** — the Maldives emits less than 0.01% of global CO_2. Its leaders have described sea level rise as an existential threat, and the Maldives has been a vocal advocate at COP summits for the 1.5°C target and for Loss and Damage funding.

Responses:

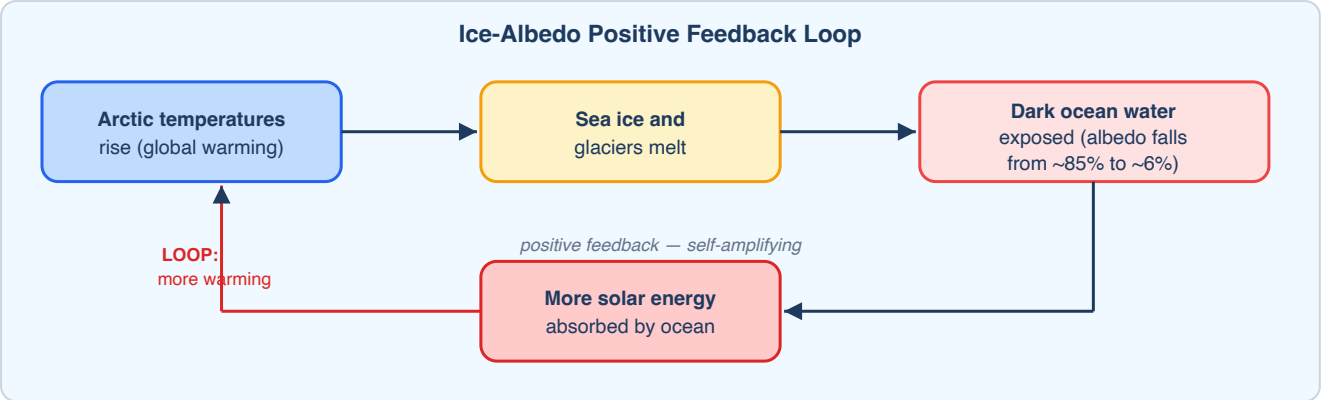
- The Maldives government has invested heavily in sea walls, storm barriers, and elevated infrastructure, but these are short-term measures that cannot address long-term inundation.
- The country aims to become carbon-neutral by 2030 — a symbolic gesture given its minimal emissions — to demonstrate international leadership.
- The Alliance of Small Island States (AOSIS) — representing 44 countries — has been a powerful voice in international climate negotiations, advocating for the 1.5°C target enshrined in the Paris Agreement.

Exam point: The Maldives and Tuvalu are powerful examples for answering questions about climate justice, the differential impact of climate change, and the gap between those responsible for emissions and those who suffer the consequences.

7. Physical Impacts — Positive Feedback Loops

Positive feedback loops amplify warming and are one of the most alarming aspects of climate science. They mean that warming triggers further warming.

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Other key positive feedback loops include:

- **Permafrost-methane feedback** — as Arctic permafrost thaws, it releases methane and CO_2 stored for millennia (the permafrost carbon pool is estimated at 1,500 billion tonnes of carbon — nearly double the current atmospheric carbon stock). This additional CO_2 and methane causes further warming, thawing more permafrost.
- **Water vapour feedback** — as temperatures rise, more water evaporates. Water vapour is itself a powerful greenhouse gas, amplifying warming. This approximately doubles the warming effect of CO_2 alone.
- **Forest dieback** — warmer temperatures and changing rainfall patterns cause forest die-off and more wildfires. Dead/burning forest releases stored carbon, reducing carbon sink capacity and accelerating warming.

8. Mitigation Strategies

Mitigation — actions that reduce the amount of GHGs being emitted, or that increase carbon absorption (carbon sinks). Aims to slow or prevent climate change from worsening. Tackles the *cause* of climate change.

Strategy	How it works	Example	Evaluation
Renewable energy	Replaces fossil fuels with solar, wind, tidal, HEP — all low/zero carbon	Germany's Energiewende ("energy transition") — over half of electricity from renewables (2023); UK has Europe's largest offshore wind	+ Rapidly falling costs (solar down 89% since 2010); + no emissions during operation; – intermittent (wind/solar

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		capacity; Costa Rica generates ~98% of its electricity from renewables in most years	need backup or storage); – land use conflicts
Energy efficiency	Reduces energy demand through better insulation, LED lighting, efficient appliances, electric vehicles	EU energy efficiency directive; UK smart meter rollout; Norway: around 90% of new car sales were electric in 2024 — enabled by hydropower grid; Japan's Top Runner programme	+ Reduces demand (not just supply-side fix); + saves money long-term; – upfront costs high for consumers; – behaviour change needed
Carbon capture and storage (CCS)	Captures CO_2 from power station exhausts and stores underground (depleted oil/gas fields, saline aquifers)	Sleipner project (Norway) — stores 1 million tonnes CO_2/year under North Sea since 1996; Drax power station (UK) trialling bio-energy CCS (BECCS)	+ Can decarbonise fossil fuel use and industry; – expensive (~$50–100/tonne CO_2); – not yet at scale globally; – permanence of storage uncertain
Reforestation and afforestation	Planting trees and protecting forests to absorb CO_2; REDD+ scheme pays LIDCs to preserve forests	Great Green Wall (Africa) — 8,000km belt across Sahel; China's afforestation (66,000 km²/year); UK planting 30,000 hectares/year target	+ Natural and relatively cheap; + biodiversity co-benefits; – takes decades to mature; – forests can burn (releasing stored carbon); – land competition with food production
Nuclear power	Generates electricity with very low lifecycle carbon emissions (~12g CO_2/kWh vs. coal ~820g)	France: 70% of electricity from nuclear — one of Europe's lowest per capita electricity carbon footprints; UK building Hinkley Point C (Somerset)	+ Reliable baseload; + very low operational carbon; – nuclear waste storage (10,000+ years); – very high cost and long construction time; – public opposition after Fukushima (2011)

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Carbon pricing	Carbon tax or cap-and-trade system makes emitting CO_2 more expensive, incentivising reduction	EU ETS (Emissions Trading System) — companies must buy permits to emit; UK ETS (since 2021); British Columbia Carbon Tax (Canada, 2008–2025, when the consumer tax was scrapped)	+ Economic incentive to reduce; + revenue can fund green investment; – political opposition; – carbon leakage (production moves to countries without pricing)

9. Adaptation Strategies

Adaptation — actions that adjust society, infrastructure, and behaviour to reduce the harmful effects of climate change that is already happening or inevitable. Accepts some level of change as unavoidable. Tackles the *effects* of climate change.

Strategy	Example	Evaluation
Sea defences (hard engineering)	**Thames Barrier** (built 1982, cost £534m): 10 rotating steel gates, spans 520m across Thames. Has been closed 200+ times since opening, far more often in recent decades than in the 1980s. It is expected to need replacing or upgrading around the 2070s — the Thames Estuary 2100 plan is preparing for this. **Netherlands Delta Works**: Maeslant storm surge barrier near Rotterdam — fully automated, one of the world's largest moving structures, cost €450m.	+ Proven technology; + protects high-value urban areas; – extremely expensive; – cannot keep pace with accelerating sea level rise indefinitely; – disrupts estuarine ecosystems
Managed retreat (soft engineering)	**Medmerry, West Sussex, UK** (2013): 183 hectares of farmland deliberately flooded to create intertidal habitat, allowing natural coastline retreat. Protects Selsey and surrounding villages more effectively than maintaining the failing shingle bank. The ~300-hectare site includes 183 hectares of new intertidal habitat.	+ Cheaper long-term than hard engineering; + creates valuable wildlife habitats; + works with natural processes; – loss of farmland and property; – community opposition; –

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		political difficulties in implementing
Drought-resistant crops	Drought-tolerant maize varieties (WEMA — Water Efficient Maize for Africa project): used by many farmers in Kenya, Uganda and Tanzania; yields are significantly higher than conventional varieties under drought conditions. Golden rice (GM crop with Vitamin A) for food and nutritional security in Asia.	+ Reduces food insecurity; + can be targeted to LIDCs; + relatively affordable once developed; – GM controversy and regulatory barriers; – intellectual property concerns (seed patents)
Water management	Drip irrigation (pioneered in Israel) — delivers water directly to roots, reducing use by 30–60%. Rainwater harvesting in India (rooftop systems, traditional johad ponds revived). Singapore's NEWater scheme (recycled water meets 40% of demand). Morocco's desalination programme.	+ Reduces demand pressure; + affordable at small scale; – requires significant infrastructure investment at scale; – desalination is energy-intensive
Urban design and heat adaptation	Green roofs (Stuttgart, Germany — city has mandatory green roofs on new flat-roofed buildings, reducing urban heat island effect by 1–2°C). Medellín, Colombia: urban green corridors reduced local temperature by 2°C. UK heatwave action plan: designating cool spaces, public health messaging, school and hospital guidelines.	+ Multiple co-benefits (biodiversity, air quality, mental health, stormwater management); + can be retrofitted; – expensive at city scale; – limited effect alone without mitigation
Early warning systems	Bangladesh cyclone warning system: established after 1991 cyclone killed 138,000 people. Cyclone Sidr (2007, similar intensity): only 3,500 deaths due to evacuation. Bangladesh has tens of thousands of trained volunteers, thousands of cyclone shelters, and an SMS/radio warning system.	+ Very cost-effective relative to lives and property saved; + replicable in other LIDCs; – requires sustained government commitment and funding; – does not address underlying causes

Exam Tip — Mitigation vs Adaptation Comparison: Mitigation tackles the *cause* (reducing GHG emissions); adaptation tackles the *effects* (adjusting to changes). Both are necessary and neither is sufficient alone. For evaluation: some argue adaptation is becoming more urgent because

current emission pledges are insufficient to prevent 2°C+ of warming, so we must adapt to changes already locked in. However, adaptation without mitigation is ultimately futile — there are limits to how much change human societies and ecosystems can adapt to. The two approaches are complementary, not alternatives.

10. International Responses and Agreements

Kyoto Protocol (1997)

Adopted at COP3 in Kyoto, Japan. Entered into force in 2005 after Russia's ratification provided the required threshold.

- **Key provisions:** The first legally binding international climate treaty. Set specific emission reduction targets for developed (Annex I) countries: an average of **5.2% below 1990 levels** by the commitment period 2008–2012. Created three "flexible mechanisms" — Emissions Trading, the Clean Development Mechanism (CDM), and Joint Implementation — to allow countries to meet targets cost-effectively.
- **Strengths:** First legally binding treaty; created an international framework; carbon markets (including the EU ETS) developed out of Kyoto mechanisms; some countries (EU) did meet targets.
- **Weaknesses:** USA signed but never ratified (Senate refused). Did not include developing nations (China, India, Brazil) — who argued their historical emissions were low and they had a right to develop. Canada withdrew in 2011. By the second commitment period (2012–2020), only 37 countries participated. Global emissions continued to rise throughout the Kyoto period. The Protocol is now widely viewed as having failed to address the scale of the problem.

Copenhagen Accord (2009)

COP15, Copenhagen, Denmark. Widely seen as a diplomatic failure. World leaders were expected to agree a binding successor to Kyoto, but negotiations collapsed. The resulting non-binding "accord" was "noted" (not formally adopted). Countries submitted voluntary pledges but the accord included no binding emission targets. However, it was the first agreement to include the principle of limiting warming to 2°C and established the concept of $100 billion per year in climate finance from rich to poor nations by 2020 — a target that was first met in 2022, two years late.

Paris Agreement (2015)

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COP21, Paris, France. Adopted by **196 parties** — the most universally agreed international climate deal. Entered into force 4 November 2016.

- **Temperature target:** Limit global warming to **well below 2°C above pre-industrial levels**, and pursue efforts to limit it to **1.5°C**. The 1.5°C target was a major diplomatic victory for small island states.
- **Nationally Determined Contributions (NDCs):** Each country sets its own emissions reduction pledges. NDCs must be submitted every five years and are expected to become progressively more ambitious ("ratchet mechanism"). NDCs are not legally binding — only the obligation to submit them is binding.
- **Climate finance:** Developed nations reaffirmed the commitment to mobilise $100 billion per year by 2020 to help LIDCs adapt and reduce emissions. At COP29 (Baku, 2024) a new goal of at least $300 billion a year by 2035 was agreed.
- **Loss and Damage:** Recognised the need to address climate impacts that exceed adaptation capacity, but the mechanism was limited.
- **Strengths:** Near-universal participation (195 signatories); regular review mechanism; 1.5°C ambition is meaningful; includes LIDCs in the framework; transparent reporting requirements.
- **Weaknesses:** NDCs are not legally binding; current collective pledges put the world on track for approximately **2.5–3°C** of warming, far exceeding the 1.5°C target (UNEP Emissions Gap Report 2023); climate finance target of $100bn often fell short; USA withdrew under President Trump (2017–2021), rejoined under Biden, and began withdrawing again in 2025; developing nations argue the burden is unfair given historical emissions.

COP26 — Glasgow (2021)

- **Glasgow Climate Pact:** Called for phasing down (not out) "unabated" coal power and phasing out "inefficient" fossil fuel subsidies — historic first mention of fossil fuels in a COP decision but language was weakened by India/China at the last moment.
- **Deforestation pledge:** Over 130 countries, covering 90% of world's forests, pledged to halt and reverse deforestation by 2030, backed by $19 billion in public and private funding.
- **Global Methane Pledge:** Over 100 countries pledged to reduce methane emissions 30% below 2020 levels by 2030 — significant given methane's high short-term warming potential.
- **Finance:** $130 trillion in private finance pledged via the Glasgow Financial Alliance for Net Zero (GFANZ) — though the quality and accountability of these pledges was widely questioned.

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- **NDC ratchet:** Countries with insufficient NDCs asked to submit stronger plans by end of 2022 — compliance was mixed.
- **Weaknesses:** "Unabated coal" loophole allows continued use with CCS. Deforestation pledge not legally binding (a similar pledge in 2014 from New York had little effect). $100bn climate finance target was again not met. Loss and Damage fund not established (agreed at COP27, operationalised at COP28).

COP28 — Dubai (2023)

- First agreement to explicitly mention "transitioning away from fossil fuels" — a historic first, though "phase-out" language was rejected by oil-producing states.
- **Loss and Damage Fund** formally established and operationalised — $700 million pledged in initial contributions (compared to estimated need of $400 billion per year).
- Tripling of renewable energy capacity globally by 2030 agreed.
- Hosted by the UAE (a major oil producer), with Sultan Al Jaber — the CEO of Abu Dhabi National Oil Company — as COP28 president, attracting significant controversy over conflict of interest.

Key Criticisms of International Climate Agreements

- **Non-binding nature** — without legal enforcement, countries can fall short without penalty. The difference between Kyoto (binding targets) and Paris (voluntary NDCs) reflects the political reality that binding agreements are hard to achieve globally.
- **Historical responsibility vs current emissions** — the USA and EU have emitted the majority of cumulative GHGs since the Industrial Revolution. China and India argue they have a right to develop economically as ACs did. China is now the world's largest annual emitter (~27%) but its per-capita emissions (~8 tonnes CO_2/year) remain below the USA (~14 tonnes CO_2/year).
- **Insufficient ambition** — UNEP Emissions Gap Report (2023): current policies put world on track for 2.5–3°C warming. A 45% emissions reduction by 2030 is needed to stay on the 1.5°C pathway. Only a handful of countries have NDCs consistent with 1.5°C.
- **Climate finance shortfall** — the $100bn/year commitment to LIDCs was not met until 2022, two years late; much of the "finance" is loans rather than grants, increasing debt burdens on vulnerable nations. Estimates of LIDC adaptation needs range from $160bn–$340bn per year by 2030.

11. UK-Specific Impacts of Climate Change

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The UK is warming at a rate consistent with global trends. The UK Climate Change Risk Assessment (CCRA3, 2022) identifies urgent risks for the UK across multiple sectors.

Flooding

- The UK's wettest years on record include 2000 and 2012, and very wet years have become more frequent. Atmospheric rivers and more intense low-pressure systems are linked to climate change.
- **Winter 2019–20 flooding** — Storm Dennis and Storm Ciara brought record rainfall to Yorkshire, Derbyshire, and Wales; the Calder Valley (West Yorkshire) flooded again and rivers such as the Wye reached record levels. Thousands of homes flooded, with damage running to hundreds of millions of pounds.
- The Environment Agency (2024) estimated about 6.3 million properties in England are in areas at risk of flooding from rivers, the sea or surface water, potentially rising to around 8 million by 2050.
- Surface water flooding (sewers overwhelmed by intense rainfall) is a growing problem in urban areas — Storm Babet (2023) caused major river and surface water flooding in eastern Scotland (e.g. Brechin).

Heatwaves

- **July 2022 heatwave** — temperatures reached **40.3°C at Coningsby, Lincolnshire** — the first time 40°C was ever recorded in the UK. The previous record was 38.7°C (2019). The Met Office had never before issued a Red Heat Warning. Around 3,000 heat-associated deaths were estimated across the summer 2022 heatwaves in England.
- UK infrastructure was ill-prepared: road surfaces buckled, railway lines warped (requiring speed restrictions), Luton Airport runway damaged. Schools were not equipped with adequate ventilation or shading.
- Climate attribution studies suggest the July 2022 temperatures were made at least **10 times more likely** by human-caused climate change.
- The Met Office projects that summers as hot as 2018 could be typical by the 2050s.

Agricultural Change

- **Potential benefits (short term):** Warmer temperatures extend the growing season by approximately 4–6 weeks for some crops. New crops becoming viable in southern England — maize, sunflowers, and certain wine grape varieties now grown commercially. The UK wine industry has

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expanded dramatically: over 900 vineyards, producing award-winning sparkling wines in Kent and Sussex.

- **Risks:** Increased summer drought threatens rain-fed agriculture, particularly in eastern England (already the driest region, receiving less rainfall than Jerusalem). Soil moisture deficits have increased since the 1970s. New pests and diseases moving northward (e.g. bluetongue virus, spread by midges — first reached the UK in 2007; the BTV-3 strain arrived in 2023). Apple orchards in Kent threatened by late spring frosts (warming winters mean trees lose frost-hardiness; late cold snaps then kill blossom).
- Oilseed rape (canola) yields have declined as cabbage stem flea beetle infestations have worsened — partly linked to warmer winters that allow beetle populations to survive at higher densities.

Coastal Change

- UK sea levels have risen by approximately **10–20cm** since 1900. Thames Estuary rising at 3.5mm/year (including land subsidence).
- East Yorkshire's Holderness coast loses an average of **2 metres per year** to erosion — the fastest-eroding coastline in Europe. Villages including Skipsea and Atwick are at severe risk within decades. The village of Mappleton spent £2 million on rock groynes in 1991 to protect itself, but accelerated erosion of neighbouring unprotected coast resulted.
- Managed retreat is increasingly recognised as necessary for low-value agricultural land along vulnerable coasts.

Biodiversity in the UK

- Many UK species have shifted their ranges northward since the 1970s, tracking warming temperatures.
- New species arriving: little egrets, ring-necked parakeets, and bee-eaters now breeding in the UK. Edible dormice spreading.
- Species declining: mountain hares, capercaillie, dotterel, and other cold-adapted upland species losing habitat as snowfall decreases in Scottish Highlands. Puffin colonies threatened by sand eel decline (sand eels moving northward as waters warm, reducing food availability for puffins).

12. Future Projections and Uncertainty

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- The IPCC projects global average temperatures could rise by roughly **1.4°C to over 4°C** above pre-industrial levels by 2100, depending on future emissions. Global temperatures in 2024 were already around 1.5°C above pre-industrial levels.
- Sea levels rose by about 20 cm between 1901 and 2018 and could rise by a further 0.3–1 m (or more) by 2100.

Why projections are uncertain

Source of uncertainty	Explanation
Future emissions	Depend on population growth, economic development, technology and government policy — no-one knows how quickly the world will cut emissions.
Natural factors	Volcanic eruptions and changes in solar output cannot be predicted far in advance.
Feedback mechanisms	Positive feedbacks (melting ice reducing albedo, thawing permafrost releasing methane) could speed up warming in ways that are hard to model.
Model limitations	Climate models simplify complex systems such as clouds and ocean circulation.

Exam Tip: "Explain why there is uncertainty about future climate change" is a common OCR question. Separate human uncertainty (emissions) from physical uncertainty (feedbacks, natural factors).

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